Turbo screening for non-negative LASSO

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***Abstract*—Many algorithms allow to tackle the LASSO problem but particular data structures can give them a hard time. In this paper, we present a new Turbo screening method combining screening tests in both primal and dual LASSO problem with the aim of improving a solving algorithm in terms of computational cost. The Turbo screening approach is generic and is particularly well suited for algorithm computing the gradient of the objective function somewhere in its iterations.**

***Index Terms*—LASSO, screening, primal screening, dual screening, safe regions, unsafe regions, gap regions, ellipsoidal regions**

I. Introduction

The *non-negative* LASSO is a famous convex problem finding applications in various fields. It can be expressed as

$$\mathbf{x}^\star \in \underset{\mathbf{x}\geq \mathbf{0}_n}{\operatorname{argmin}} \left\{ \mathrm{P}(\mathbf{x}) = \tfrac{1}{2}\left\|\mathbf{y}-\mathbf{A}\mathbf{x}\right\|_2^2 + \lambda \mathbf{1}_n^{\mathrm{T}}\mathbf{x} \right\} \qquad (\mathcal{P})$$

for a given *dictionary* $\mathbf{A} \in \mathbb{R}^{m\times n}$, a given *observation* $\mathbf{y} \in \mathbb{R}^m$ and a given *regularization parameter* $\lambda \geq 0$. It seeks for a sparse vector $\mathbf{x}^\star$ approximating $\mathbf{y}$ through the coefficients of $\mathbf{A}$. The larger λ, the sparser $\mathbf{x}^\star$. Many algorithms can tackle this problem very efficiently. Bests known are FISTA [Cite] and ADMM [Cite] . However, they face their limits in some cases. When the dimension of $\mathbf{A}$ is too large, iterations can be costly to handle. Furthermore, algorithms usually struggle to converge when a high accuracy is required. Recently, *screening tests* [Cite] were developed to face the dimension issue by reducing iteratively the size of the problem during the resolution process. In this paper, we introduce new screening tests to tackle ($\mathcal{P}$) when a high accuracy is required which allow to bypass the last iterations where methods are usually slow to converge.

A. Problem derivations

In the following, we assume that the columns $\mathbf{a}_i$ of $\mathbf{A}$ and $\mathbf{y}$ are normalized. ($\mathcal{P}$) creates a sparse solution and once have $\|\mathbf{x}^\star\|_0 \leq m$ for any $\lambda \geq 0$. For $\lambda \geq \lambda_{\max} := \max_i \mathbf{a}_i^{\mathrm{T}}\mathbf{y}$, $\mathbf{x}^\star$ is identically null. The ratio $0 \leq \lambda/\lambda_{\max} \leq 1$ is often used to tune the regularization amount targeted in $\mathbf{x}^\star$. The dual problem of ($\mathcal{P}$) can be parametrized as

$$\mathbf{u}^\star = \underset{\mathbf{u}\in\mathcal{U}}{\operatorname{argmax}} \left\{ \mathrm{D}(\mathbf{u}) = \tfrac{1}{2}\left\|\mathbf{y}\right\|_2^2 - \tfrac{\lambda^2}{2}\left\|\tfrac{\mathbf{y}}{\lambda}-\mathbf{u}\right\|_2^2 \right\} \qquad (\mathcal{D})$$

where $\mathcal{U} = \{\mathbf{u} \in \mathbb{R}^m, \mathbf{A}^{\mathrm{T}}\mathbf{u} \leq \mathbf{1}\}$. Even tough $\mathbf{u}^\star = \operatorname{proj}_{\mathcal{U}}(\mathbf{y}/\lambda)$ has a simple expression, it is as hard to compute as $\mathbf{x}^\star$.

In the following, $\mathbf{x}_{\mathrm{f}}$ and $\mathbf{u}_{\mathrm{f}}$ denotes respectively primal and dual feasible points and $\operatorname{gap}(\mathbf{x}_{\mathrm{f}}, \mathbf{u}_{\mathrm{f}})$ denotes the duality gap. Such primal and dual feasible points can be created from non-feasible points using the following *scaling operations* :

$$\mathbf{x}_{\mathrm{f}} = \max\{\mathbf{0}, \mathbf{x}\} \tag{1a}$$

$$\mathbf{u}_{\mathrm{f}} = \begin{cases} \mathbf{u} & \text{if } \max_i \mathbf{a}_i^{\mathrm{T}}\mathbf{u} \leq 1 \\ \frac{\mathbf{u}}{\max_i \mathbf{a}_i^{\mathrm{T}}\mathbf{u}} & \text{otherwise} \end{cases} \tag{1b}$$

For ($\mathcal{P}$), stationarity and complementary slackness optimality conditions are

$$\mathbf{a}_i^{\mathrm{T}}\mathbf{u}^\star \in \begin{cases} \{1\} & \text{if } x_i^\star > 0 \\ ]-\infty, 1] & \text{if } x_i^\star = 0 \end{cases} \tag{2a}$$

$$\mathbf{A}\mathbf{x}^\star = \mathbf{y} - \lambda\mathbf{u}^\star \tag{2b}$$

Furthermore, from ($\mathcal{P}$) and ($\mathcal{D}$) one can define the *primal-active set* and the *dual-active set* as

$$\mathcal{I}_{\mathrm{p}}^\star = \{i \in [\![1,n]\!], x_i^\star = 0\} \quad \mathcal{I}_{\mathrm{d}}^\star = \{i \in [\![1,n]\!], \mathbf{a}_i^{\mathrm{T}}\mathbf{u}^\star = 1\}$$

In the following, we assume that $\mathcal{I}_{\mathrm{p}}^\star \cap \mathcal{I}_{\mathrm{d}}^\star = \emptyset$. When $\mathbf{A}$ is sampled randomly, $\mathbb{P}(\mathcal{I}_{\mathrm{p}}^\star \cap \mathcal{I}_{\mathrm{d}}^\star \neq \emptyset) = 0$, regardless of the generation law. Under this assumption, one can derive from (2a) that

$$x_i^\star > 0 \iff \mathbf{a}_i^{\mathrm{T}}\mathbf{u}^\star = 1 \tag{3a}$$

$$x_i^\star = 0 \iff \mathbf{a}_i^{\mathrm{T}}\mathbf{u}^\star \in]-\infty, 1[\tag{3b}$$

Suppose that $\mathcal{I}_{\mathrm{p}}^\star$ (or equivalently $\mathcal{I}_{\mathrm{d}}^\star$) is known and let $\tilde{\mathbf{A}} = [\mathbf{a}_i]_{i\in\mathcal{I}_{\mathrm{d}}^\star}$ and $\tilde{\mathbf{x}} = [x_i]_{i\in\mathcal{I}_{\mathrm{d}}^\star}$. We have from (2a) and (2b) that

$$\tilde{\mathbf{A}}^{\mathrm{T}}\tilde{\mathbf{A}}\tilde{\mathbf{x}}^\star = \tilde{\mathbf{A}}^{\mathrm{T}}\mathbf{y} - \lambda\mathbf{1} \tag{4}$$

The dimension of (4) is relatively small in the classical use-cases of ($\mathcal{P}$) because usually $|\mathcal{I}_{\mathrm{d}}^\star| << m$. We can notice that contrary to (2b), $\mathbf{u}^\star$ is not involved. The solution of ($\mathcal{P}$) can be reconstructed from $\tilde{\mathbf{x}}^\star$ as

$$\begin{cases} x_i^\star = \tilde{x}_i^\star & \text{if } i \in \mathcal{I}_{\mathrm{d}}^\star \\ x_i^\star = 0 & \text{if } i \in \mathcal{I}_{\mathrm{p}}^\star \end{cases} \tag{5}$$

An ideal way to tackle ($\mathcal{P}$) is to solve (4) and use (5) to get $\mathbf{x}^\star$ but in practice, no prior information about $\mathcal{I}_{\mathrm{p}}^\star$ or $\mathcal{I}_{\mathrm{d}}^\star$ is known.

B. Outline of the method

Screening tests [Cite] , hereafter called *primal screening tests*, were developed the last decade in order to improve

an algorithm by identifying some of the components of $\mathcal{I}_{\mathrm{p}}^{\star}$. This allows to remove the corresponding column of $\mathbf{A}$ and coefficient of $\mathbf{x}$ during the resolution process. The computational cost of each iteration is therefore reduced during the algorithm. Our aim was to find a way to know when $\mathcal{I}_{\mathrm{p}}^{\star}$ has been fully identified in order to directly solve (4) and recover the solution with (5) with a high accuracy. Knowing when $\mathcal{I}_{\mathrm{p}}^{\star}$ is fully known allows to stop the algorithm sooner and therefore to bypass the last iterations where methods have usually difficulties to reach a high precision. We developed new *dual screening tests* allowing to identify components of $\mathcal{I}_{\mathrm{d}}^{\star}$. When all the coefficients have been identified as belonging to $\mathcal{I}_{\mathrm{p}}^{\star}$ or $\mathcal{I}_{\mathrm{d}}^{\star}$, one know that these sets are fully identified. One can thus switch to the resolution of (4) and use (5) to recover $\mathbf{x}^{\star}$ very accurately.

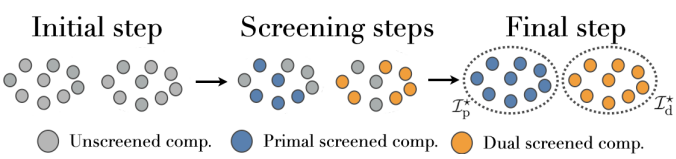

Fig. 1. Each grey element is a column of $\mathbf{A}$ or equivalently a coefficient of $\mathbf{x}^{\star}$. In the final step, we are sure that $\mathcal{I}_{\mathrm{p}}^{\star}$ is fully known and we can switch to the resolution of (4).

For a given problem, we want to diminish the computational cost to reach a high precision for a method by overloading it with both primal and dual screening tests. We called this approach *Turbo screening*. The challenge is not to over-penalize the method when adding these screening tests so we always seek for cheap ways to proceed them. One will see that a method computing $\nabla \mathrm{P}$ somewhere in its iterations is particularly well suited for the Turbo screening approach.

We first review literature about primal screening tests in II and we present new dual screening tests in III. A generic Turbo screening algorithm and numerical results are presented in IV and V.

II. Primal screening tests

Primal screening tests aim to identify elements of $\mathcal{I}_{\mathrm{p}}^{\star}$. Assume that a region $\mathcal{R}$ bounding $\mathbf{u}^{\star}$ is known, we can deduce from (3b) that

$$\max_{\mathbf{u}\in\mathcal{R}} \ \mathbf{a}_i^{\mathrm{T}}\mathbf{u} < 1 \Longrightarrow i \in \mathcal{I}_{\mathrm{p}}^{\star} \tag{6}$$

A good choice for $\mathcal{R}$ is a region cheap to construct, with a small size and making (6) easy to solve. The tighter the region, the better the screening rate. Typically, a spherical region with a size decreasing with the dual gap is a good region. The *primal gap region* [Cite] satisfies these criteria and can be constructed using the inequality

$$\|\mathbf{u}^{\star} - \mathbf{u}_{\mathrm{f}}\|_2 \leq \sqrt{2\mathrm{gap}(\mathbf{x}_{\mathrm{f}}, \mathbf{u}_{\mathrm{f}})}/\lambda \tag{7}$$

which holds for any $\mathbf{x}_{\mathrm{f}}$ and $\mathbf{u}_{\mathrm{f}}$. Successive points are provided by the method and can be used to create $\mathbf{x}_{\mathrm{f}}$ and $\mathbf{u}_{\mathrm{f}}$ with (1a), (2b) and (1b). The spherical shape allows to solve (6) easily. Usually, $\mathrm{gap}(\mathbf{x}_{\mathrm{f}}, \mathbf{u}_{\mathrm{f}}) \to 0$ during a method so the region is tightened at each iteration and more and more elements of $\mathcal{I}_{\mathrm{p}}^{\star}$ can be identified. Below a certain dual gap, this set can be fully identified.

The main advantage of primal screening tests using the primal gap region is that in most cases, they can be proceeded at a low computational cost. Indeed, if the values $\mathbf{A}\mathbf{x}^k$, $\mathbf{u}^k := \mathbf{y} - \mathbf{A}\mathbf{x}^k$ and $\mathbf{A}^{\mathrm{T}}\mathbf{u}^k$ are already computed at the iteration k, the test can be done in $o(m+n)$ multiplications. These values can be extracted from the computation of $\nabla \mathrm{P}(\mathbf{x}^k)$, which is done in almost all methods allowing to solve ($\mathcal{P}$). As the computation of $\nabla \mathrm{P}(\mathbf{x}^k)$ costs at least $o(mn)$ multiplications, the additional cost of primal screening tests does not impact a lot the method performances.

III. Dual screening tests

Dual screening tests are constructed in the same way as primal screening tests. In contrast to them, dual screening tests do not allow to reduce the complexity of the method iterations in general. Thus, we want these tests even cheaper to proceed.

A. General dual screening test

Let a region $\mathcal{R}$ bounding $\mathbf{x}^{\star}$, we can deduce from (3a) that

$$\min_{\mathbf{x}\in\mathcal{R}} x_i > 0 \Longrightarrow i \in \mathcal{I}_{\mathrm{d}}^{\star} \tag{8}$$

As with primal screening tests, we want $\mathcal{R}$ to be simple to construct, to have a simple shape and a small size. This makes (8) simple to solve and allows to have a good dual-screening rate without penalizing too much the method.

B. Some safe regions for dual screening

Once can use the following spherical and ellipsoidal regions for dual screening tests.

1) Dual gap region: If $n \leq m$ and given $\mathbf{x}_{\mathrm{f}}$ and $\mathbf{u}_{\mathrm{f}}$, one have

$$\|\mathbf{x}^{\star} - \mathbf{x}_{\mathrm{f}}\|_2 \leq \sqrt{2\mathrm{gap}(\mathbf{x}_{\mathrm{f}}, \mathbf{u}_{\mathrm{f}})}/\alpha \tag{9}$$

for $0 \leq \alpha \leq \lambda_{\min}(\mathbf{A}^{\mathrm{T}}\mathbf{A})$ where $\lambda_{\min}(\mathbf{X})$ is the smallest eigenvalue of $\mathbf{X}$. The finest dual gap region is created with $\alpha = \lambda_{\min}(\mathbf{A}^{\mathrm{T}}\mathbf{A})$. For a spherical region $\mathcal{S}(\mathbf{c}, r)$, the solution of (6) is $\min_{\mathbf{x}\in\mathcal{S}(\mathbf{c},r)} x_i = \mathbf{c}_i - r$.

2) Dual ellipsoidal region: If $n \leq m$ and given $\mathbf{x}_{\mathrm{f}}$ and $\mathbf{u}_{\mathrm{f}}$, one have

$$(\mathbf{x}^{\star} - \mathbf{x}_{\mathrm{f}})^{\mathrm{T}}(\mathbf{A}^{\mathrm{T}}\mathbf{A})(\mathbf{x}^{\star} - \mathbf{x}_{\mathrm{f}}) \leq 2\mathrm{gap}(\mathbf{x}_{\mathrm{f}}, \mathbf{u}_{\mathrm{f}}) \tag{10}$$

Thus, we can create an ellipsoidal dual screening region with center $\mathbf{x}_{\mathrm{f}}$ and covariance matrix $2\mathrm{gap}(\mathbf{x}_{\mathrm{f}}, \mathbf{u}_{\mathrm{f}})(\mathbf{A}^{\mathrm{T}}\mathbf{A})^{-1}$. The solution of (6) with this region is given in [Cite] .

The two regions derived above have a size decreasing with $\mathrm{gap}(\mathbf{x}_{\mathrm{f}}, \mathbf{u}_{\mathrm{f}})$. Therefore, using dual screening tests in a method where $\mathrm{gap}(\mathbf{x}_{\mathrm{f}}, \mathbf{u}_{\mathrm{f}}) \to 0$ ensure that below a certain dual gap, $\mathcal{I}_{\mathrm{d}}^{\star}$ can be fully identified. Likewise for the primal gap sphere, the value of $\mathrm{gap}(\mathbf{x}_{\mathrm{f}}, \mathbf{u}_{\mathrm{f}})$ can be computed cheaply if $\nabla \mathrm{P}(\mathbf{x}^k)$ has already been computed. Notice that with these regions, dual screening can only begin once $n \leq m$ (*i.e.* the dimension has been enough reduced by primal screening tests).

C. Dual screening in practice

Even if $\mathrm{gap}(\mathbf{x}_\mathrm{f}, \mathbf{u}_\mathrm{f})$ is usually cheap to compute, we have to know a feasible value for α in (9) or the value of $(\mathbf{A}^\mathrm{T}\mathbf{A})^{-1}$ in (10) to proceed dual screening tests. Usually, such values are unknown in methods allowing to solve ($\mathcal{P}$) efficiently. Computing these values can over-penalize the method, leading ultimately to a higher computational cost with Turbo screening than without. We propose an heuristic to address cases where the computation of α or $(\mathbf{A}^\mathrm{T}\mathbf{A})^{-1}$ is too costly.

A region is called 'safe' [Cite] when it allows to dual-screen without mistakes. We have observed that in practice, the dual gap region (9) doesn't produces lot of screening mistakes when its radius is relaxed by setting $\alpha \geq \lambda_{\min}(\mathbf{A}^\mathrm{T}\mathbf{A})$, even if theoretically the region is unsafe. The idea is to find a value for α which is easy to compute and close to $\lambda_{\min}(\mathbf{A}^\mathrm{T}\mathbf{A})$. In this way, we can identify iteratively $\mathcal{I}_\mathrm{d}^\star$, possibly with mistakes. As the region is unsafe, we continue to proceed safe primal screening tests to correct possible mistakes of unsafe dual screening tests. Every time new components of $\mathcal{I}_\mathrm{d}^\star$ are identified, we can solve (4) and use optimality conditions for this problem to find out if $\mathcal{I}_\mathrm{d}^\star$ is indeed well identified. In this way, our heuristic may be akin to a greedy Active-set heuristic [Cite] . To efficiently solve (4), the value of $(\tilde{\mathbf{A}}^\mathrm{T}\tilde{\mathbf{A}})^{-1}$ can be computed once and rank-one updates rules [Cite] can be use every time components of $\mathcal{I}_\mathrm{d}^\star$ are added or removed. As usually, $|\mathcal{I}_\mathrm{d}^\star| << m$, the additional computational cost of this heuristic is marginal comparing to the iterations of most solving methods for ($\mathcal{P}$). Furthermore, primal screening tests tend to eliminate tricky coefficients which may have been dual screened wrongly.

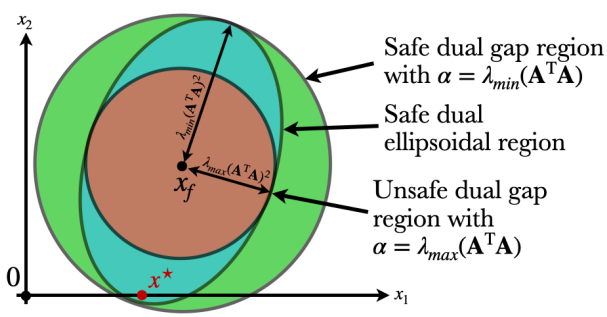

Fig. 2. Safe and unsafe dual screening regions. $x_2^\star$ is safely dual-screened by the three regions. $x_1^\star$ is also wrongly dual-screened by the red region.

IV. Turbo screening algorithm

Turbo screening approach is tested with FISTA [Cite] and with the Frank-Wolfe (FW) algorithm [Cite] . We use the primal gap sphere and the dual gap sphere with a relaxed radius for the screening tests. With FISTA, we can set $\alpha = L$ where L is the Lipschitz constant used in the iterations. In FW, we can for example set $\alpha = \min(\|\mathbf{A}\|_1, \|\mathbf{A}\|_\infty)$. The value of $\nabla\mathrm{P}$ is already computed within the iterations of FISTA and FW. The scheme of a method using Turbo screening is the following :

Algorithm 1: Turbo screening

Input: $\mathbf{x}^0$, $\mathbf{A}$, $\mathbf{y}$, λ

$\mathcal{I}_\mathrm{p}^\star \leftarrow \emptyset$, $\mathcal{I}_\mathrm{d}^\star \leftarrow \emptyset$
while *stopping criteria not met* **do**
  $\mathbf{x}, \nabla\mathrm{P} \leftarrow$ `MethodIteration`$(\mathbf{x}, \mathbf{A}, \mathbf{y}, \lambda)$
  $\mathcal{I}_\mathrm{p}^\star \leftarrow \mathcal{I}_\mathrm{p}^\star \cup$ `PrimalScreening`$(\nabla\mathrm{P}, \lambda)$
  $\mathbf{x}, \mathbf{A} \leftarrow$ `ReduceDimension`$(\mathcal{I}_\mathrm{p}^\star)$
  if $n \leq m$ **then**
    $\alpha \leftarrow$ `FindRelaxedValue()`
    $\mathcal{I}_\mathrm{d}^\star \leftarrow \mathcal{I}_\mathrm{d}^\star \cup$ `DualScreening`$(\nabla\mathrm{P}, \alpha)$
    $(\tilde{\mathbf{A}}^\mathrm{T}\tilde{\mathbf{A}})^{-1} \leftarrow$ `RankOneUpdate`$(\mathcal{I}_\mathrm{d}^\star)$
    $\tilde{\mathbf{x}}^\star \leftarrow$ `SolveReducedProblem`$((\tilde{\mathbf{A}}^\mathrm{T}\tilde{\mathbf{A}})^{-1})$
    if $\mathcal{I}_\mathrm{d}^\star$ *is well identified* **then**
      return $\mathbf{x}^\star$ using (5)
    end
  end
end

A continuation method [Cite] can be used to find a good starting point. To efficiently proceed screening tests, the value of $\nabla\mathrm{P}$ must be re-used to compute $\mathrm{gap}(\mathbf{x}_\mathrm{f}, \mathbf{u}_\mathrm{f})$ cheaply. As a relaxed value for α can also be computed cheaply, screening tests do not over-penalize the method. To determine if $\mathcal{I}_\mathrm{d}^\star$ is well identified, one can use optimality conditions for (4) which can be checked with a marginal cost because $|\mathcal{I}_\mathrm{d}^\star|$ is usually small.

V. Numerical results

In this final section, we show how Turbo screening improves FISTA and FW. The target precision on the duality gap is 10^{-15}. We use backtracking rules in FISTA [Cite] . For FW, we use the improved version used for the LASSO described in [Cite] . Dictionaries named 'gaussian' and 'uniform' have columns generated according to the corresponding probability law. The Gaussian dictionary has a low correlation between its columns, contrary to the uniform one. We only count the number of multiplications to evaluate the computational cost of a method as additional have a marginal computation cost in practice.

A. Safeness of the unsafe dual gap sphere

In the algorithm 1, we use an unsafe region for the dual screening test. Therefore, it is possible that some coefficients are dual-screened wrongly. However, we mentioned that in practice, the region is not often mistaken. The following table quantify the region error for two dictionaries and a range of regularization. We can notice that usually, there is less than 1.5% of the coefficients which are wrongly dual-screened so the relaxed dual gap region used is not 'too unsafe'.

B. Classical convergence scheme

Usually, the improvement of Turbo screening is more important when $m << n$. In such cases, $|\mathcal{I}_\mathrm{d}^\star|$ is low so the greedy active set strategy is efficient to determine its components.

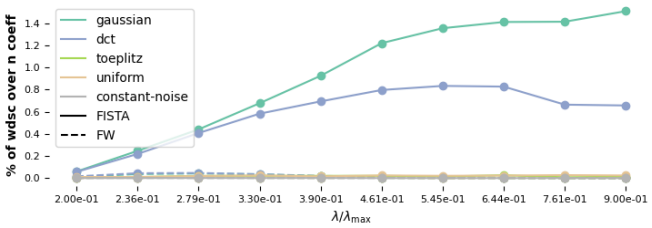

Fig. 3. Percentage of mistakes over the n coefficient of $\mathbf{x}$ during FISTA and FW for different dictionaries and regularizations.

Once this set is well identified, the solving of (4) allows to recover the optimal solution with a machine-epsilon accuracy. When only primal screening tests are used, the method still has to iterate to reach a high accuracy but the last iterations do not improve a lot the accuracy. In cases where $\lambda/\lambda_{\max}$ is low or when there is lot of correlation between columns of $\mathbf{A}$, methods are usually slow to converge. Thus, $\mathrm{gap}(\mathbf{x}_\mathrm{f}, \mathbf{u}_\mathrm{f})$ created using the points outputted by it stays large and the screening rate is not good. In such cases, Turbo screening is less efficient. This highlights the fact that the Turbo screening approach is very dependent on the performance of the solving method used.

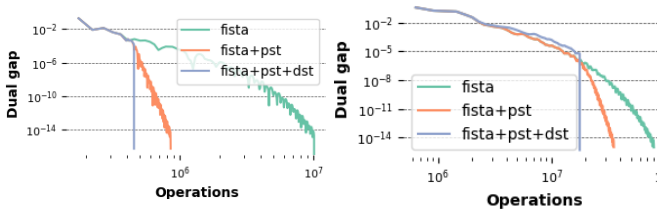

Fig. 4. FISTA without screening, with primal screening and with turbo (primal-dual) screening. Left : 50×300 Gaussian dictionary with $\lambda/\lambda_{\max} = 0.6$. Right : 300×300 Gaussian dictionary with $\lambda/\lambda_{\max} = 0.2$.

C. Gain using Turbo screening

In the following, a gain equal to 2 means that Turbo screening needs twice less operations than the method with primal screening only to met the desired accuracy. All the gains greater than 5 have been capped but for FW, the gain goes up to 200. As explained previously, Turbo screening is efficient when the dictionary is very flat ($m << n$) because $|\mathcal{I}_\mathrm{d}^\star|$ is small. We can also notice that there is an upturn in performance when the dictionary is tall ($n \leq m$) and that is because dual screening tests can be proceeded from the top of the algorithm. Thus, the identification of $\mathcal{I}_\mathrm{d}^\star$ is very aggressive but more mistakes are made by the relaxed dual gap region.

Conclusion

Starting from existing primal screening tests, we create new dual screening tests allowing to identify the active set of the dual LASSO problem. Even though these new tests do not allow to reduce the problem complexity likewise primal screening tests, we have presented an heuristic acting as an Active set method. By proceeding both primal and dual screening tests, we simply solve a linear system allowing to bypass the last iterations of most methods which usually struggle when a high accuracy is needed.

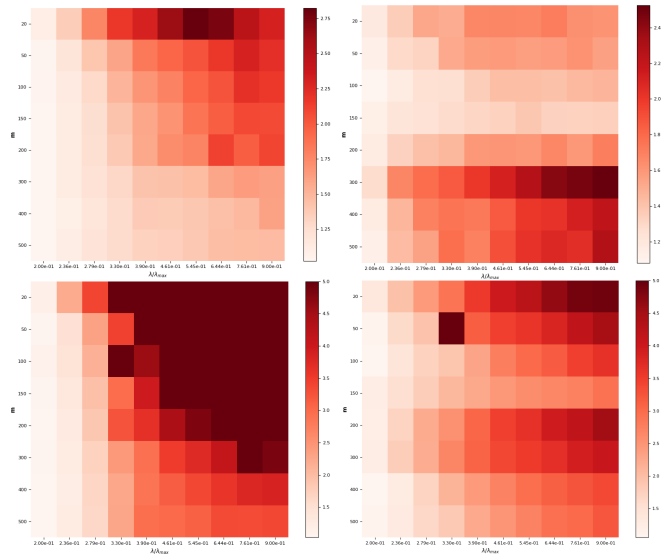

Fig. 5. Turbo screening gain against method with primal screening only with $n = 300$ fixed. Top row : FISTA. Bottom row : FW. Left column : Gaussian dictionaries. Right column : Uniform dictionaries.

Acknowledgment

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References

[1] G. Eason, B. Noble, and I. N. Sneddon, "On certain integrals of Lipschitz-Hankel type involving products of Bessel functions," Phil. Trans. Roy. Soc. London, vol. A247, pp. 529–551, April 1955.
[2] G. Eason, B. Noble, and I. N. Sneddon, "On certain integrals of Lipschitz-Hankel type involving products of Bessel functions," Phil. Trans. Roy. Soc. London, vol. A247, pp. 529–551, April 1955.
[3] G. Eason, B. Noble, and I. N. Sneddon, "On certain integrals of Lipschitz-Hankel type involving products of Bessel functions," Phil. Trans. Roy. Soc. London, vol. A247, pp. 529–551, April 1955.
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[5] G. Eason, B. Noble, and I. N. Sneddon, "On certain integrals of Lipschitz-Hankel type involving products of Bessel functions," Phil. Trans. Roy. Soc. London, vol. A247, pp. 529–551, April 1955.
[6] G. Eason, B. Noble, and I. N. Sneddon, "On certain integrals of Lipschitz-Hankel type involving products of Bessel functions," Phil. Trans. Roy. Soc. London, vol. A247, pp. 529–551, April 1955.
[7] G. Eason, B. Noble, and I. N. Sneddon, "On certain integrals of Lipschitz-Hankel type involving products of Bessel functions," Phil. Trans. Roy. Soc. London, vol. A247, pp. 529–551, April 1955.
[8] G. Eason, B. Noble, and I. N. Sneddon, "On certain integrals of Lipschitz-Hankel type involving products of Bessel functions," Phil. Trans. Roy. Soc. London, vol. A247, pp. 529–551, April 1955.